

Bike Demand Prediction — Results

Random Forest model results from the project.

Model comparison

min_samples_leaf	MAE
1	46.04
2	46.30
5	47.87
10	51.30

Lowest validation MAE: 46.04 with min_samples_leaf = 1.

Sample of actual vs predicted values

```
▶ comparison=pd.DataFrame({  
    "Actual":val_y,  
    "Predicted":model1Predict  
})  
  
comparison.head(10)
```

...

Actual Predicted



13034	398	468.64
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13035	568	616.60
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13036	295	271.76
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13037	174	174.56
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13038	201	191.86
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13039	257	268.51
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13040	236	217.12
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13041	230	219.23
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Highest average absolute error by hour

```
error_by_hour = (  
    comparison.groupby("hr")["Absolute_Error"].mean().sort_values(ascending=False)  
)  
  
print(error_by_hour.head(5))
```

```
***  hr  
     17    110.743352  
     18     99.776538  
      8     92.580157  
     19     87.725330  
     16     80.456978  
     Name: Absolute_Error, dtype: float64
```

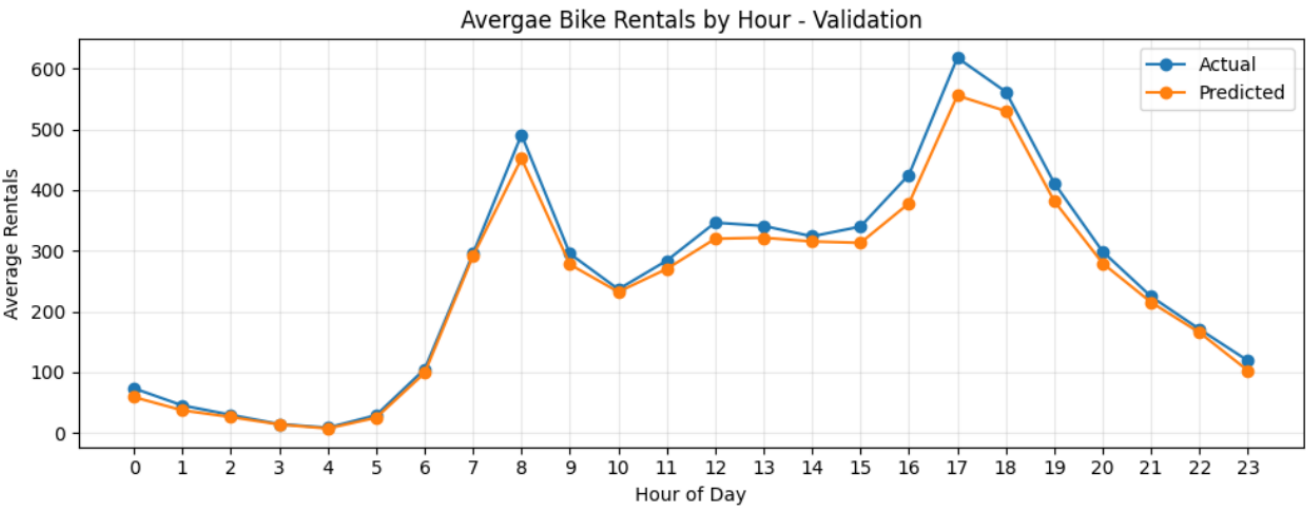
Hourly summary

```
hour_summary=comparison.groupby("hr")[["Actual","Predicted","Absolute_Error"]].mean()  
  
print(hour_summary.sort_values("Absolute_Error",ascending=False).head(5).round(2))
```

```
...      Actual  Predicted  Absolute_Error  
hr  
17  618.08     555.76         110.74  
18  562.01     530.21          99.78  
8   491.34     452.14          92.58  
19  411.56     381.97          87.73  
16  425.31     378.26          80.46
```

Average bike rentals by hour — validation

...



Missing values and duplicate checks

... Missing Values:

instant	0
dteday	0
season	0
yr	0
mnth	0
hr	0
holiday	0
weekday	0
workingday	0
weathersit	0
temp	0
atemp	0
hum	0
windspeed	0
casual	0
registered	0
cnt	0
dtype:	int64

Duplicate rows: 0

Duplicate date-hour pairs: 0